ZCAS University

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

MID-SEMESTER EXAM

OBJECT ORIENTED PROGRAMMING (CCS 2121)

INSTRUCTIONS:

1. This paper consist of two Sections, A, B and C
2. **You are required to answer all questions in Sections A and B as well as any two questions from Section C**
3. This paper carries a total of 100 marks
4. Neat work with clear examples is high recommended.
5. Candidates must not turn this page until the invigilator tells them to do so.
6. Time allowed is 1hr 30 Minutes

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ection A (Multiple choice) 20 Marks

1. Which of the following data types is used to store whole numbers in Java?
   A) double B) float C) int D) char
2. What is the difference between a class and an object in Java?
   A) A class is an instance of an object. B) An object is an instance of a class. C) Classes and objects are the same thing. D) Classes and objects are not used in Java.
3. Which of the following is a valid identifier in Java? A) 1var B) var1 C) var 1 D) var-one
4. Which of the following is not a primitive data type in Java? A) boolean B) int C) array D) double
5. Which of the following control structures in Java allows a block of code to be executed repeatedly?
   A) if-else statement B) for loop C) switch statement D) do-while loop
6. Which of the following is a method declaration in Java? A) public int myMethod() {} B) public myMethod(int x) { } C) public void myMethod() {} D) public static myMethod() {}
7. Which of the following is a valid Java expression? A) (5 > 4) && (3 < 4) B) (5 > 4) || (3 < 4) C) (5 > 4) ||| (3 < 4) D) (5 > 4) &&& (3 < 4)
8. Which of the following is not a valid Java operator? A) + B) % C) ! D) @
9. What is the purpose of a constructor in Java? A) To declare variables. B) To initialize objects. C) To define methods. D) To declare classes.
10. Which of the following is a reference data type in Java? A) boolean B) int C) double D) String
11. What is the difference between a private and public method in Java?
    A) A private method can be accessed from any class. B) A public method can only be accessed within its class.
    C) A private method can only be accessed within its class.
    D) A public method cannot be accessed within its class.
12. Which of the following is a valid Java class definition? A) class My_Class {}
    B) class myClass {} C) class MyClass {} D) class my-class {}
13. Which of the following statements declares and initializes a variable of the int data type in Java?
    A) int x = 5; B) x = 5; C) int x; D) x;
14. Which of the following is not a comparison operator in Java? A) > B) < C) == D) +
15. Which of the following is a valid Java comment? A) // This is a comment B) # This is a comment C) /* This is a comment */ D) <!-- This is a comment -->
16. What is the difference between a primitive data type and a reference data type in Java? A). Primitive data types are objects, while reference data types are not. B). Primitive data types are stored on the stack, while reference data types are stored on the heap. C). Primitive data types are passed by reference, while reference data types are passed by value. D). Primitive data types are built-in to Java, while reference data types must be defined by the programmer.
17. Which of the following is a valid Java identifier? A). 3hello B). hello_3 C). hello-world D). Hello World
18. Which of the following is NOT a primitive data type in Java? A). int B). boolean C). string D). double
19. What is the syntax for declaring a variable in Java? A). type name; B). name type; C). name = value; D). value name;
20. What is the difference between a class and an object in Java?A). A class is a blueprint for an object, while an object is an instance of a class. B). A class is an object, while an object is a method. C). A class is a method, while an object is a blueprint for a class. D). A class is a variable, while an object is a data type.

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Section B Filling the Blanks (40 Marks)

1. A _____ is a named storage location in memory used to store data.
2. Java has _____ primitive data types.
3. A _____ data type is a type that refers to an object in memory.
4. A _____ data type is a built-in type that is not an object.
5. An _____ is a unique identifier for a variable, method, or class in Java.
6. An identifier in Java must start with a _____, _____, or _____.
7. A _____ is a blueprint for an object in Java.
8. An _____ is an instance of a class in Java.
9. A _____ is a special method that is used to create an object in Java.
10. A _____ is a set of instructions that performs a specific task in Java.
11. In Java, a method belongs to a _____.
12. In Java, a function does not belong to a _____.
13. The _____ keyword is used to declare a variable in Java.
14. A _____ is a group of related classes and interfaces in Java.
15. An _____ statement is used to include a class or package in Java.
16. A _____ structure is a block of code that controls the flow of a program in Java.
17. The three basic types of control structures in Java are _____, _____, and _____.
18. A _____ is a control structure that allows code to be executed repeatedly in Java.
19. An _____ statement is a control structure that allows code to be executed conditionally in Java.
20. An _____ statement is used to check a condition in Java.

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ection C: Coding (Write any two of the following) 40 Marks

1. Write a Java program that defines a class called Circle. The class should have an instance variable for the radius and two methods, getArea() and getCircumference(), that calculate the area and circumference of the circle. Use the following formulas:
   *Area = π * radius^2 Circumference = 2 * π * radius*
2. Write a Java program that defines a class called Car. The class should have instance variables for the make, model, and year, and a method called printInfo() that prints out the car's information in the following format:
   Make: [make] Model: [model] Year: [year]
3. Write a Java program that uses a while loop to print out the numbers 1 through 5.
4. Write a Java program that defines a class called Dog. The class should have instance variables for the name, breed, and age, and a method called bark() that prints out a message saying "Woof!"

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